

Searching for Axion-Like Particles at FCC-ee: Projected Sensitivity in the Invisible and Prompt-Resolved Channels at the Z Pole

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Abstract

We present projected sensitivity estimates for the production of axion-like particles (ALPs) at the proposed Future Circular Collider operating at the Z pole ($\sqrt{s} = 91.2 \text{ GeV}$, $\mathcal{L} = 150 \text{ ab}^{-1}$) via the associated production process $e^+e^- \rightarrow \gamma a$. Two detector signatures are analyzed: the *invisible* channel, in which the ALP escapes the detector ($\ell_a > L_{\text{max}}$), and the *prompt-resolved* channel, in which the ALP decays to two separated photons inside the tracker ($\ell_a < L_{\text{min}}$, $\Delta\theta > \Delta\theta_{\text{res}}$). Signal events are generated with MADGRAPH5_AMC@NLO, processed through PYTHIA 8 for ALP decay and lifetime, and passed through a DELPHES fast simulation of the IDEA detector concept. SM backgrounds for both channels are simulated, binned, and normalized to the full target luminosity. Projected 90% CL exclusion contours are derived using a binned Asimov $\Delta\chi^2$ method, with analytic production and lifetime formulas corrected by detector-level signal scans. For the invisible channel, the analysis reaches $g_{a\gamma\gamma} \approx 5.5 \times 10^{-7} \text{ GeV}^{-1}$ over the ALP mass range 0.01–0.92 GeV. The prompt-resolved channel extends sensitivity to $g_{a\gamma\gamma} \sim 10^{-5} \text{--} 3 \times 10^{-4} \text{ GeV}^{-1}$ for $m_a \in [0.61, 80] \text{ GeV}$. Both projected contours surpass current collider bounds by one to two orders of magnitude in coupling at comparable masses. The analysis machinery is validated against the published Belle II ALP exclusion boundary, reproducing it to within $|\log_{10}(g_{\text{closure}}/g_{\text{published}})| \leq 7.6 \times 10^{-3}$. Additional validation gates check the MadGraph cross section, the ALP width and Pythia lifetime convention, the Delphes output, and the reconstructed ALP invariant-mass or recoil-energy observable at every detector-level signal point.

1 Introduction

The Peccei–Quinn (PQ) mechanism, proposed to explain the absence of strong CP violation [1–3], predicts the existence of a pseudo-Nambu–Goldstone boson, the axion. The simplest requirement for the PQ mechanism to resolve the strong CP problem constrains the axion mass and its couplings to a narrow band in the coupling–mass plane—the so-called QCD axion line. However, extensions of this framework that decouple the mass from the coupling give rise to *axion-like particles* (ALPs): pseudo-scalars that couple to Standard Model (SM) particles with arbitrary mass–coupling combinations [4–6].

The ALP–photon coupling,

$$\mathcal{L} \supset -\frac{1}{4}g_{a\gamma\gamma} a F_{\mu\nu} \tilde{F}^{\mu\nu}, \quad (1)$$

is particularly well-motivated. It arises generically from the anomalous coupling of any axion-like field to the electromagnetic field strength $F_{\mu\nu}$, and is bounded across many orders of magnitude in mass by astrophysical, cosmological, and laboratory experiments [12]. In the mass range 10 MeV–80 GeV relevant to collider searches, the most stringent direct constraints come from e^+e^- collider experiments (LEP, Belle II, BaBar, BESIII) and beam-dump experiments, and leave significant regions of parameter space unexplored at couplings $g_{a\gamma\gamma} \lesssim 10^{-4} \text{ GeV}^{-1}$.

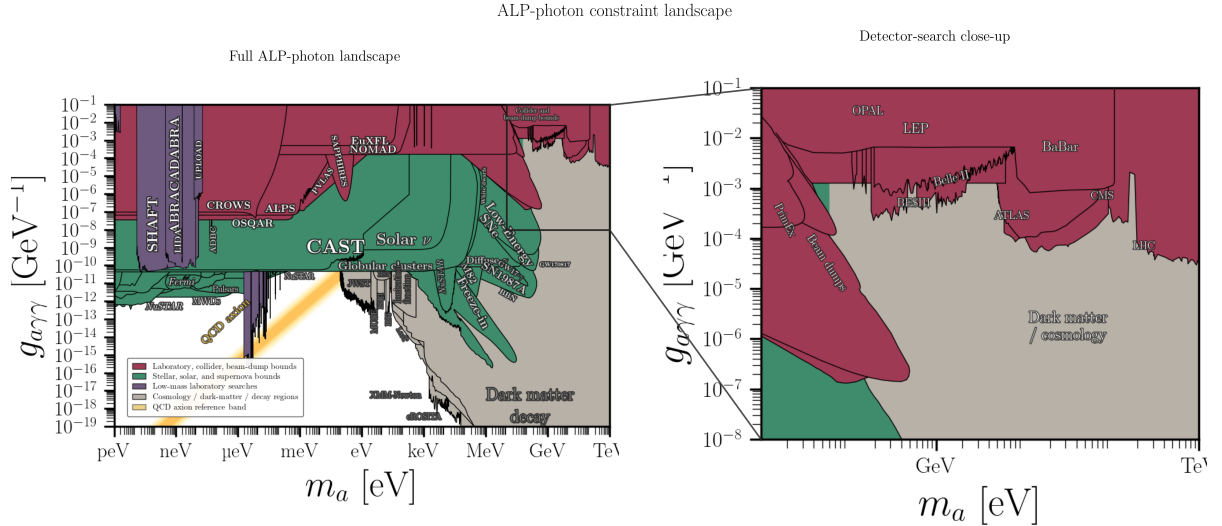


Figure 1: Current ALP-photon constraint landscape used as context for this project, produced from the pinned AxionLimits repository. The left panel shows the broad landscape, while the right panel zooms into the collider-relevant mass range. Solid colored regions show existing laboratory, astrophysical, cosmological, and dark-matter-motivated constraints or reference regions. The FCC-ee projections are deliberately omitted here and are introduced later as the project result.

The Future Circular Collider (FCC-ee) is a proposed e^+e^- machine that would operate at several center-of-mass energies from the Z pole ($\sqrt{s} \approx 91.2 \text{ GeV}$) up to the $t\bar{t}$ thresh-

old [8, 9]. At the Z pole, the baseline luminosity target of 150 ab^{-1} —corresponding to approximately 5×10^{12} Z bosons—provides an extraordinarily clean and prolific environment for precision measurements and rare-process searches. For photophilic ALPs, the associated production cross section $\sigma(e^+e^- \rightarrow \gamma a) \propto g_{a\gamma\gamma}^2(1 - m_a^2/s)^3$ grows quadratically with the coupling and benefits enormously from the high luminosity, making FCC-ee a uniquely powerful probe. The Z-pole run is chosen as the baseline deliverable for three practical and physics reasons. First, it is the FCC-ee stage with the largest expected integrated luminosity, so it gives the strongest statistics-driven reach for the single-coupling process studied here. Second, the fixed center-of-mass energy gives a simple mono-energetic recoil photon, $E_\gamma = (s - m_a^2)/(2\sqrt{s})$, making the invisible and prompt-resolved analyses especially transparent. Third, the Z-pole overlaps the mass range where existing collider constraints from Belle II, BaBar, BESIII, LEP, and the LHC leave open regions that can be tested in a clean e^+e^- environment. Other FCC-ee stages are scientifically interesting, but each stage has a different luminosity, background composition, recoil-energy range, and detector acceptance optimization. They are therefore treated as extensions rather than part of the controlled final deliverable.

This paper presents projected ALP sensitivity estimates at the FCC-ee Z pole, focusing on the two experimentally cleanest signatures: the invisible channel (ALP escapes) and the prompt-resolved channel (ALP decays to two separated photons). The analysis uses a complete simulation chain from matrix-element generation to fast detector response, validated against published experimental results.

A central goal of the project is not only to draw an ALP sensitivity curve, but to build a reusable ALP simulation and analysis pipeline. This matters because an ALP search is not determined by the production formula alone. The same point in the $(m_a, g_{a\gamma\gamma})$ plane can look like missing energy, three photons, two merged photon clusters, or a displaced decay depending on the ALP lifetime and boost. The detector response then decides how many of those ideal events are actually reconstructed, while Standard Model processes decide how difficult the signal is to distinguish from ordinary events. A pipeline that connects theory, event generation, detector simulation, background modeling, and limit setting is therefore needed if the final projection is to mean more than an idealized rate estimate. It also makes the analysis extensible: the same structure can later be reused for other FCC-ee energies, different detector assumptions, displaced or merged channels, or additional ALP operators.

Paper structure. Section 2 reviews the relevant ALP physics. Section 3 describes the simulation and detector modeling. Section 4 presents the signal classification, background estimation, and limit-setting method. Section 5 reports the projected exclusion contours. Section 6 discusses the physics reach, limitations, and outlook.

2 Theory

2.1 Analytic model for photophilic ALPs

This analysis assumes a single effective operator, Eq. (1), with all direct ALP couplings to fermions, gluons, W bosons, and Z bosons set to zero at tree level. The only model parameters scanned are therefore the ALP mass m_a and the photon coupling $g_{a\gamma\gamma}$. To calculate any process we need to compute the amplitude of the interaction. This is derived from the Lagrangian of the interaction (1). In this benchmark the total width is taken to be

$$\Gamma_{\text{tot}} \simeq \Gamma(a \rightarrow \gamma\gamma), \quad (2)$$

because $a \rightarrow \gamma\gamma$ is the only open tree-level decay channel in the implemented model. Loop-induced fermion modes are neglected consistently with the photophilic benchmark and with the UFO parameter choice used in the simulation.

2.2 Associated-production kinematics and amplitude

In the center-of-mass frame, the total four-momentum of the collision is $P^\mu = p_{e^-}^\mu + p_{e^+}^\mu = (\sqrt{s}, 0, 0, 0)$, where $\sqrt{s}$ is the center-of-mass energy of the collision, and p_{e^-} and p_{e^+} are the incoming electron and positron four-momenta. Momentum conservation tells us that this is equal to the total four-momentum of the outgoing ALP and photon, $P^\mu = p_a^\mu + k^\mu$, where p_a is the outgoing ALP four-momentum and k is the outgoing recoil-photon four-momentum. We then have $P^\mu - k^\mu = p_a^\mu$, so squaring it gives

$$\begin{aligned} (P^\mu - k^\mu)(P_\mu - k_\mu) &= p_a^\mu p_{a\mu} \\ P^2 - 2P \cdot k + k^2 &= p_a^2 \\ s - 2P \cdot k &= m_a^2 \\ s - 2(\sqrt{s}, \mathbf{0}) \cdot (E_\gamma, \mathbf{k}) &= m_a^2 \\ s - 2\sqrt{s}E_\gamma &= m_a^2. \end{aligned} \quad (3)$$

Here $k^2 = 0$ for the recoil photon, since photons are massless, and $p_a^2 = m_a^2$ for the ALP. So in the center-of-mass frame we can express the recoil photon energy as

$$E_{\gamma, \text{recoil}} = \frac{s - m_a^2}{2\sqrt{s}}. \quad (4)$$

It follows that the ALP energy and the three-momentum magnitude are

$$E_a = \sqrt{s} - E_\gamma = \frac{s + m_a^2}{2\sqrt{s}}, \quad |\mathbf{p}_a| = |\mathbf{k}| = \frac{s - m_a^2}{2\sqrt{s}}. \quad (5)$$

106 $E_{\gamma,\text{recoil}}$ is therefore one of the main signals we use, since we can reconstruct the invariant
 107 mass of the ALP when we detect the recoil photon. At fixed $\sqrt{s}$ and m_a , a two-body final
 108 state has no freedom to choose a different recoil-photon energy before detector smearing
 109 or initial-state radiation. The ALP is boosted with respect to its rest frame, and we can
 110 get the ALP boost from these kinematic variables as

$$\gamma_a = \frac{E_a}{m_a}, \quad \beta_a \gamma_a = \frac{|\mathbf{p}_a|}{m_a}. \quad (6)$$

111 The cross section is the quantity that tells us how often the process of an ALP being
 112 created via associated production in an e^+e^- collision occurs for a given beam luminosity,
 113 $N = \sigma \mathcal{L}$. We can calculate $\sigma(e^+e^- \rightarrow a\gamma)$ by using Fermi's golden rule in relativistic
 114 phase-space form:

$$d\sigma = \frac{S}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} |\mathcal{M}|^2 (2\pi)^4 \delta^{(4)}\left(p_1 + p_2 - \sum_f p_f\right) \prod_f \frac{d^3 \mathbf{p}_f}{(2\pi)^3 2E_f}. \quad (7)$$

115 Here p_1 and p_2 are the incoming electron and positron momenta, and m_1 and m_2 are their
 116 masses. The factor with the square root is the incoming flux. The delta function enforces
 117 conservation of the total four-momentum, and the factor S is a symmetry factor that
 118 avoids double counting identical particles in the final state. For $e^+e^- \rightarrow \gamma a$ the photon
 119 and ALP are different particles, so $S = 1$. The final product is the Lorentz-invariant
 120 phase space available to the outgoing particles.

121 To compute any process, we first need its amplitude, which contains all of the dynamical
 122 information. We obtain it by evaluating the relevant Feynman diagram. The first
 123 vertex of the interaction is the basic QED current: the incoming electron and positron
 124 annihilate into an off-shell photon, $e^+e^- \rightarrow \gamma^*(q)$, where $q = p_{e^-} + p_{e^+}$ and $q^2 = s$. This
 125 virtual photon then produces the recoil photon and the ALP through the ALP-photon
 126 interaction. The full associated-production process is therefore

$$e^+e^- \rightarrow \gamma a. \quad (8)$$

127 From the coupling between the ALP and photons in Eq. (1), we can derive the ALP-
 128 photon vertex

$$V^{\mu\nu}(k_1, k_2) = -ig_{a\gamma\gamma} \epsilon^{\mu\nu\alpha\beta} k_{1,\alpha} k_{2,\beta}, \quad (9)$$

129 where $\epsilon^{\mu\nu\alpha\beta}$ is the four-dimensional Levi-Civita symbol. It is a completely antisymmetric
 130 object which encodes the CP-odd nature of the ALP-photon interaction. Both photon
 131 four-momenta, k_1^α and k_2^β , are taken incoming to the vertex. However, in production,
 132 one of these photon legs is the virtual γ^* , so we designate its four-momentum as q ; and
 133 the other is the recoil photon γ_{recoil} , which is outgoing from the ALP vertex, so the all-

incoming momentum assigned to that leg is $-k$. The ALP-photon vertex is then

$$V^{\nu\rho}(q, -k) = -ig_{a\gamma\gamma}\epsilon^{\nu\rho\alpha\beta}q_\alpha(-k)_\beta = ig_{a\gamma\gamma}\epsilon^{\nu\rho\alpha\beta}q_\alpha k_\beta. \quad (10)$$

This is why the vertex appears with a plus sign in the production amplitude.

The amplitude for $e^+e^- \rightarrow a\gamma$ is therefore obtained by multiplying all the pieces of associated production: the ordinary QED current, the virtual-photon propagator, the ALP-photon vertex, and the polarization vector of the outgoing photon,

$$\begin{aligned} \mathcal{M} &= i \underbrace{\bar{v}(p_{e^+})(-ie\gamma^\mu)u(p_{e^-})}_{\substack{e^+e^-\gamma^* \\ \text{current}}} \underbrace{\left(\frac{-ig_{\mu\nu}}{s}\right)}_{\substack{\gamma^* \\ \text{propagator}}} \underbrace{(ig_{a\gamma\gamma}\epsilon^{\nu\rho\alpha\beta}q_\alpha k_\beta)_{\gamma^*\gamma a}}_{\text{vertex}} \underbrace{\varepsilon_\rho^*(k)}_{\substack{\text{outgoing } \gamma \\ \text{polarization}}} \\ &= \frac{eg_{a\gamma\gamma}}{s} [\bar{v}(p_{e^+})\gamma^\mu u(p_{e^-})] \epsilon_{\mu\nu\alpha\beta} q^\alpha k^\beta \varepsilon^{*\nu}(k). \end{aligned} \quad (11)$$

Here we contracted the Lorentz indices with the metric and the overall phase has been dropped, since only $|\mathcal{M}|^2$ enters the cross section. The electric charge e comes from the ordinary QED vertex where the electron-positron pair annihilates into a virtual photon. The spinors $u(p_{e^-})$ and $\bar{v}(p_{e^+})$ describe the incoming electron and positron, and γ^μ is a Dirac matrix. The factor $1/s$ is the propagator of the intermediate virtual photon after contracting with the photon-current indices. The factor $g_{a\gamma\gamma}\epsilon_{\mu\nu\alpha\beta}q^\alpha k^\beta$ is the ALP-photon vertex from Eq. (10). Finally, $\varepsilon^{*\nu}(k)$ is the polarization vector of the observed outgoing photon, since we have to specify the physical polarization state of the recoil photon. Because the detector does not measure photon polarization, when calculating the cross section we sum over the allowed photon polarizations.

We now want to square the amplitude. A useful way to do this is to separate the electron current from the ALP-photon tensor, so we define

$$J^\mu = \bar{v}(p_{e^+})\gamma^\mu u(p_{e^-}), \quad A_{\mu\nu} = \epsilon_{\mu\nu\alpha\beta} q^\alpha k^\beta. \quad (12)$$

The amplitude then looks like

$$\mathcal{M} = \frac{eg_{a\gamma\gamma}}{s} J^\mu A_{\mu\nu} \varepsilon^{*\nu}(k). \quad (13)$$

For a fixed incoming electron spin, incoming positron spin, and outgoing photon polarization, the squared amplitude is

$$|\mathcal{M}|^2 = \frac{e^2 g_{a\gamma\gamma}^2}{s^2} J^\mu J^{\mu'*} A_{\mu\nu} A_{\mu'\nu'} \varepsilon^{*\nu}(k) \varepsilon^{\nu'}(k). \quad (14)$$

In the FCC-ee collision, the beams are unpolarized, so we do not know which initial spin state was present. We therefore average over the two electron spin states and the

156 two positron spin states, giving a factor $1/4$. The detector also does not measure the
 157 polarization of the final photon, so we sum over the two physical final photon polarizations:

$$\overline{|\mathcal{M}|^2} = \frac{e^2 g_{a\gamma\gamma}^2}{s^2} \left[\frac{1}{4} \sum_{s_-, s_+} J^\mu J^{\mu'*} \right] A_{\mu\nu} A_{\mu'\nu'} \left[\sum_{\lambda} \varepsilon_{(\lambda)}^{*\nu}(k) \varepsilon_{(\lambda)}^{\nu'}(k) \right]. \quad (15)$$

158 The object in the first bracket is the spin-averaged electron tensor, which can be evaluated
 159 with the standard Dirac trace identities:

$$\begin{aligned} L^{\mu\mu'} &\equiv \frac{1}{4} \sum_{s_-, s_+} J^\mu J^{\mu'*} \\ &= \frac{1}{4} \text{Tr} \left[(p_{e^+} \cdot \gamma) \gamma^\mu (p_{e^-} \cdot \gamma) \gamma^{\mu'} \right] \\ &= p_{e^+}^\mu p_{e^-}^{\mu'} + p_{e^+}^{\mu'} p_{e^-}^\mu - g^{\mu\mu'} (p_{e^+} \cdot p_{e^-}). \end{aligned} \quad (16)$$

160 This is the standard QED trace result in the limit where $m_e^2 \ll s$. The photon polarization
 161 sum in the second square bracket can be replaced by $-g^{\nu\nu'}$, and gauge-dependent terms
 162 vanish because the amplitude is gauge invariant. With these two results Eq. (15) becomes

$$\overline{|\mathcal{M}|^2} = \frac{e^2 g_{a\gamma\gamma}^2}{s^2} L^{\mu\mu'} A_{\mu\nu} A_{\mu'\nu'} (-g^{\nu\nu'}). \quad (17)$$

163 In the center-of-mass frame we can choose the incoming electron to define the z -axis:

$$\begin{aligned} p_{e^-}^\mu &= \frac{\sqrt{s}}{2} (1, 0, 0, 1), & p_{e^+}^\mu &= \frac{\sqrt{s}}{2} (1, 0, 0, -1), \\ q^\mu &= (\sqrt{s}, 0, 0, 0), & k^\mu &= E_\gamma (1, \sin \theta_{\text{CM}}, 0, \cos \theta_{\text{CM}}). \end{aligned} \quad (18)$$

164 Substituting these momenta into Eqs. (17) and (16) gives

$$\overline{|\mathcal{M}|^2} = e^2 g_{a\gamma\gamma}^2 E_\gamma^2 (1 + \cos^2 \theta_{\text{CM}}). \quad (19)$$

165 And finally, using $E_\gamma = (\sqrt{s}/2)(1 - m_a^2/s)$ gives us

$$\overline{|\mathcal{M}|^2} = \frac{e^2 g_{a\gamma\gamma}^2 s}{4} (1 + \cos^2 \theta_{\text{CM}}) \left(1 - \frac{m_a^2}{s} \right)^2, \quad (20)$$

166 where θ_{CM} is the recoil-photon polar angle relative to the incoming electron direction. The
 167 factor $(1 + \cos^2 \theta_{\text{CM}})$ gives the angular shape, while $(1 - m_a^2/s)^2$ shows the momentum
 168 dependence of the pseudoscalar vertex. Plugging this squared amplitude into Fermi's
 169 golden rule (Eq. (7)), we now reduce the two-body phase space and obtain the differential
 170 cross section.

For associated production the final-state momenta are

$$p_3 = k \quad \text{and} \quad p_4 = p_a, \quad (21)$$

with $k^2 = 0$ for the photon and $p_a^2 = m_a^2$ for the ALP. The electron mass is negligible compared with the collider energy, so in the center-of-mass frame $m_1 = m_2 \simeq 0$ and

$$4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2} = 4|\mathbf{p}_i|\sqrt{s}. \quad (22)$$

The two-body phase-space part of Eq. (7) is

$$d\Phi_2 = (2\pi)^4 \delta^{(4)}(p_1 + p_2 - k - p_a) \frac{d^3\mathbf{k}}{(2\pi)^3 2E_\gamma} \frac{d^3\mathbf{p}_a}{(2\pi)^3 2E_a}. \quad (23)$$

In the center-of-mass frame the spatial delta function sets the two final momenta to be back-to-back, $\mathbf{p}_a = -\mathbf{k}$. The remaining energy delta function fixes the magnitude of the final momentum to the value already found from two-body kinematics,

$$|\mathbf{p}_f| = |\mathbf{k}| = \frac{s - m_a^2}{2\sqrt{s}}. \quad (24)$$

After using the delta functions, the two-body phase space becomes

$$d\Phi_2 = \frac{1}{16\pi^2} \frac{|\mathbf{p}_f|}{\sqrt{s}} d\Omega, \quad (25)$$

where $d\Omega$ is the solid-angle element of the outgoing photon. Plugging Eqs. (22) and (25) into Eq. (7) gives the standard two-body formula for unpolarized $2 \rightarrow 2$ scattering:

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\overline{\mathcal{M}}|^2, \quad (26)$$

Here $|\mathbf{p}_i| = \sqrt{s}/2$ is the magnitude of the incoming electron or positron three-momentum, and $|\mathbf{p}_f|$ is the magnitude of either final-state momentum. The ratio $|\mathbf{p}_f|/|\mathbf{p}_i|$ is the two-body phase-space suppression: if m_a approaches $\sqrt{s}$, there is little momentum left for the final state. Since

$$\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} = 1 - \frac{m_a^2}{s}, \quad (27)$$

substituting Eq. (20) into Eq. (26) gives

$$\begin{aligned} \frac{d\sigma}{d\Omega} &= \frac{1}{64\pi^2 s} \left(1 - \frac{m_a^2}{s}\right) \frac{e^2 g_{a\gamma\gamma}^2 s}{4} (1 + \cos^2 \theta_{\text{CM}}) \left(1 - \frac{m_a^2}{s}\right)^2 \\ &= \frac{e^2 g_{a\gamma\gamma}^2}{256\pi^2} (1 + \cos^2 \theta_{\text{CM}}) \left(1 - \frac{m_a^2}{s}\right)^3. \end{aligned} \quad (28)$$

186 There is no dependence on the azimuthal angle ϕ because the initial state is rotationally
 187 symmetric around the beam axis. Therefore $d\Omega = d\phi d\cos\theta_{\text{CM}}$ and

$$\frac{d\sigma}{d\cos\theta_{\text{CM}}} = \int_0^{2\pi} \frac{d\sigma}{d\Omega} d\phi = 2\pi \frac{d\sigma}{d\Omega}. \quad (29)$$

188 Using $e^2 = 4\pi\alpha$ then gives

$$\frac{d\sigma}{d\cos\theta_{\text{CM}}} = \frac{\alpha g_{a\gamma\gamma}^2}{32} (1 + \cos^2\theta_{\text{CM}}) \left(1 - \frac{m_a^2}{s}\right)^3. \quad (30)$$

189 The third power of $(1 - m_a^2/s)$ has a useful interpretation: one power comes from the
 190 available two-body phase space and two powers come from the pseudoscalar momentum
 191 dependence of the $a\gamma\gamma$ vertex. Finally,

$$\int_{-1}^1 (1 + \cos^2\theta) d\cos\theta = \frac{8}{3}, \quad (31)$$

192 so the total cross section over the full angular range is

$$\sigma(e^+e^- \rightarrow \gamma a) = \frac{\alpha g_{a\gamma\gamma}^2}{12} \left(1 - \frac{m_a^2}{s}\right)^3. \quad (32)$$

193 At $\sqrt{s} = 91.2 \text{ GeV}$ and $g_{a\gamma\gamma} = 10^{-3} \text{ GeV}^{-1}$, Eq. (32) gives $\sigma \approx 5.5 \times 10^{-5} \text{ pb}$; at $g_{a\gamma\gamma} =$
 194 10^{-6} GeV^{-1} it falls to $5.5 \times 10^{-11} \text{ pb}$. The enormous FCC-ee Z-pole luminosity ($150 \text{ ab}^{-1} =$
 195 $1.5 \times 10^8 \text{ pb}^{-1}$) compensates for small couplings and is what makes sensitivity below
 196 10^{-6} GeV^{-1} possible.

197 2.3 Diphoton Decay

198 The main decay channel for a photophilic ALP is $a \rightarrow \gamma\gamma$ (Fig. 2). In the center-of-
 199 mass frame of the ALP it decays into two back-to-back photons with equal and opposite
 200 momentum, which is our second signal.

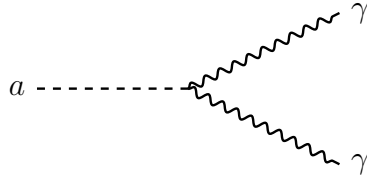


Figure 2: ALP decay to two photons, $a \rightarrow \gamma\gamma$.

201 Since the center-of-mass frame of the ALP is boosted with respect to the lab frame,
 202 the two photons do not generally have an angular separation $\Delta\theta_{\gamma\gamma} = \pi$ in the detector.
 203 Instead, the opening angle is a function of the ALP boost. Since calorimeters do not have

arbitrarily small angular resolution, if the angular separation of the photons is smaller than the calorimeter resolution, the detector can register the two photons as a single photon-like electromagnetic shower.

The vertex for the two-photon interaction is the same ALP-photon vertex used in production,

$$V^{\mu\nu}(k_1, k_2) = -ig_{a\gamma\gamma}\epsilon^{\mu\nu\alpha\beta}k_{1\alpha}k_{2\beta}. \quad (33)$$

Here k_1 and k_2 are the four-momenta of the two decay photons. Both photons are outgoing, so the all-incoming vertex in Eq. (9) is evaluated with momenta $-k_1$ and $-k_2$. The two minus signs cancel inside the product $k_{1\alpha}k_{2\beta}$, so the vertex has the same momentum dependence as before, up to an overall phase. The decay amplitude is obtained by contracting the two photon indices of the vertex with the polarization vectors of the two outgoing photons:

$$\begin{aligned} \mathcal{M} &= -iV^{\mu\nu}\epsilon_{1\mu}^*(k_1)\epsilon_{2\nu}^*(k_2) \\ &= -g_{a\gamma\gamma}\epsilon^{\mu\nu\alpha\beta}k_{1\alpha}k_{2\beta}\epsilon_{1\mu}^*\epsilon_{2\nu}^*. \end{aligned} \quad (34)$$

The star denotes complex conjugation, as appropriate for outgoing external particles. The polarization vectors ϵ_1^μ and ϵ_2^ν describe the two physical transverse polarization states of the photons. They should not be confused with the Levi-Civita symbol $\epsilon^{\mu\nu\alpha\beta}$.

To compute a decay rate, we need the squared amplitude summed over the final photon polarizations. It is useful to isolate the antisymmetric tensor carried by the ALP-photon vertex,

$$B_{\mu\nu} = \epsilon_{\mu\nu\alpha\beta}k_1^\alpha k_2^\beta, \quad \mathcal{M} = -g_{a\gamma\gamma}B_{\mu\nu}\epsilon_1^{*\mu}\epsilon_2^{*\nu}. \quad (35)$$

The detector does not measure photon polarizations, so the observable decay probability must include all allowed final polarization states. For a real photon the physical polarizations are transverse, $k \cdot \epsilon^{(\lambda)} = 0$, and the gauge-dependent terms in the polarization sum vanish in this gauge-invariant amplitude. We can therefore use

$$\sum_{\lambda} \epsilon_{\mu}^{(\lambda)*}(k)\epsilon_{\mu'}^{(\lambda)}(k) \longrightarrow -g_{\mu\mu'}. \quad (36)$$

Applying this once for each outgoing photon gives

$$\begin{aligned} \sum_{\text{pol}} |\mathcal{M}|^2 &= g_{a\gamma\gamma}^2 B_{\mu\nu}B_{\mu'\nu'}(-g^{\mu\mu'})(-g^{\nu\nu'}) \\ &= g_{a\gamma\gamma}^2 \epsilon^{\mu\nu\alpha\beta}\epsilon_{\mu\nu}^{\alpha'\beta'}k_{1\alpha}k_{2\beta}k_{1\alpha'}k_{2\beta'}. \end{aligned} \quad (37)$$

The remaining contraction uses the standard Levi-Civita identity

$$\epsilon^{\mu\nu\alpha\beta}\epsilon_{\mu\nu}^{\alpha'\beta'} = -2 \left(g^{\alpha\alpha'}g^{\beta\beta'} - g^{\alpha\beta'}g^{\beta\alpha'} \right). \quad (38)$$

227 Substituting this identity into Eq. (37) gives

$$\sum_{\text{pol}} |\mathcal{M}|^2 = -2g_{a\gamma\gamma}^2 [(k_1 \cdot k_1)(k_2 \cdot k_2) - (k_1 \cdot k_2)^2]. \quad (39)$$

228 The photons are on shell and massless, so $k_1^2 = k_2^2 = 0$. Thus the first term vanishes, and

$$\sum_{\text{pol}} |\mathcal{M}|^2 = 2g_{a\gamma\gamma}^2 (k_1 \cdot k_2)^2. \quad (40)$$

229 Momentum conservation gives $p = k_1 + k_2$, and the ALP is on shell with $p^2 = m_a^2$.

230 Therefore

$$m_a^2 = (k_1 + k_2)^2 = k_1^2 + 2k_1 \cdot k_2 + k_2^2 = 2k_1 \cdot k_2, \quad (41)$$

231 or $k_1 \cdot k_2 = m_a^2/2$. Substituting this into Eq. (40) gives

$$\sum_{\text{pol}} |\mathcal{M}|^2 = \frac{g_{a\gamma\gamma}^2 m_a^4}{2}. \quad (42)$$

232 There is no initial-state spin average in the decay because the ALP is a spin-zero
233 particle. For a two-body decay into identical massless particles,

$$\Gamma = \frac{1}{2m_a} \frac{1}{2!} \int d\Phi_2 \sum_{\text{pol}} |\mathcal{M}|^2, \quad \int d\Phi_2 = \frac{1}{8\pi}. \quad (43)$$

234 The factor $1/(2m_a)$ is the normalization for one decaying particle of mass m_a . The factor
235 $1/2!$ is needed because the two final photons are identical: without it, the same physical
236 final state would be counted twice when the labels 1 and 2 are exchanged. Substituting
237 Eq. (42) gives

$$\begin{aligned} \Gamma(a \rightarrow \gamma\gamma) &= \frac{1}{2m_a} \frac{1}{2} \frac{1}{8\pi} \left(\frac{g_{a\gamma\gamma}^2 m_a^4}{2} \right) \\ &= \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}. \end{aligned} \quad (44)$$

238 This is the width used throughout the project:

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}. \quad (45)$$

239 For correctly paired photons from the ALP decay,

$$M_{\gamma\gamma}^2 = (k_1 + k_2)^2 = 2E_1 E_2 (1 - \cos \theta_{12}) = m_a^2. \quad (46)$$

Here E_1 and E_2 are the measured photon energies and θ_{12} is the opening angle between the two photons. This invariant mass is independent of the lab-frame boost of the ALP, so correctly paired decay photons form a peak at $M_{\gamma\gamma} = m_a$. That peak is the main analysis observable in the prompt-resolved channel.

The proper lifetime and proper decay length are

$$\tau_a = \frac{1}{\Gamma_a}, \quad c\tau_a = \frac{\hbar c}{\Gamma_a}, \quad (47)$$

with $\hbar c = 1.973269804 \times 10^{-16}$ GeV m. In the lab frame the decay length is increased by the ALP boost,

$$\ell_a = \beta_a \gamma_a c\tau_a = \frac{|\mathbf{p}_a|}{m_a} \frac{\hbar c}{\Gamma(a \rightarrow \gamma\gamma)}. \quad (48)$$

where $\beta_a = v_a/c$, $\gamma_a = E_a/m_a$, and $\beta_a \gamma_a = |\mathbf{p}_a|/m_a$. Equations (45) and (48) show that ℓ_a decreases with increasing $g_{a\gamma\gamma}$ and increasing m_a : heavy, strongly coupled ALPs decay promptly, while light, weakly coupled ALPs are long-lived and escape the detector.

The probability density for an ALP to decay after travelling a lab-frame distance L is exponential,

$$f(L) = \frac{1}{\ell_a} e^{-L/\ell_a}. \quad (49)$$

Integrating this density over the relevant detector length intervals gives the probabilities used in the analytic scan:

$$P(L > L_{\max}) = e^{-L_{\max}/\ell_a}, \quad (50)$$

$$P(L < L_{\min}) = 1 - e^{-L_{\min}/\ell_a}, \quad (51)$$

$$P(L_{\min} < L < L_{\max}) = e^{-L_{\min}/\ell_a} - e^{-L_{\max}/\ell_a}. \quad (52)$$

The first line is the survival probability to the outer detector radius $L_{\max}$ and defines the invisible signature. The second is the probability to decay before the prompt-distance scale $L_{\min}$. The third is the probability to decay between those two scales, producing a displaced signature.

The angular resolution boundary follows from boosting the isotropic $a \rightarrow \gamma\gamma$ decay. For an ALP moving along the z axis, the two photons are emitted back-to-back in the ALP rest frame. The smallest lab-frame opening angle occurs when the rest-frame photons are emitted transverse to the boost, for which

$$\Delta\theta_{\gamma\gamma}^{\min} = 2 \arctan\left(\frac{1}{\gamma_a \beta_a}\right) = 2 \arctan\left(\frac{m_a}{|\mathbf{p}_a|}\right). \quad (53)$$

The physical reason is simple: a highly boosted ALP carries the two decay photons forward in almost the same direction, so the two electromagnetic showers can merge in

the calorimeter. For light ALPs, $|\mathbf{p}_a| \simeq \sqrt{s}/2$ and $\gamma_a \simeq \sqrt{s}/(2m_a)$, so

$$\Delta\theta_{\gamma\gamma}^{\min} \simeq \frac{2}{\gamma_a} \simeq \frac{4m_a}{\sqrt{s}}. \quad (54)$$

The prompt-resolved channel requires this opening angle to be larger than the detector angular resolution, $\Delta\theta_{\text{res}}$. If $\Delta\theta_{\gamma\gamma}^{\min} < \Delta\theta_{\text{res}}$, the two photons are treated as merged even if the ALP decays promptly.

2.4 Detector-level signal categories

The ALP phenomenology at the detector level is therefore determined by two parameters: the decay length ℓ_a and the minimum diphoton opening angle $\Delta\theta_{\gamma\gamma}^{\min}$ of the ALP decay products. We can therefore classify each point $(m_a, g_{a\gamma\gamma})$ in parameter space into four exclusive signatures:

1. **Invisible** ($\ell_a > L_{\text{max}}$): The ALP escapes the detector before decaying, leaving only a single photon γ_{recoil} with missing energy as signature, $e^+e^- \rightarrow \gamma_{\text{recoil}} + \cancel{E}$.
2. **Prompt-resolved** ($\ell_a < L_{\text{min}}$ **and** $\Delta\theta_{\gamma\gamma}^{\min} \geq \Delta\theta_{\text{res}}$): The ALP decays immediately into two resolvable photons, $e^+e^- \rightarrow \gamma_{\text{recoil}} + \gamma\gamma$. The three-photon final state carries an invariant-mass peak at m_a .
3. **Displaced-resolved** ($L_{\text{min}} \leq \ell_a \leq L_{\text{max}}$): The ALP decays inside the detector but away from the interaction point.
4. **Merged** ($\ell_a < L_{\text{max}}$ **and** $\Delta\theta_{\gamma\gamma}^{\min} < \Delta\theta_{\text{res}}$): The decay photons cannot be angularly resolved, so the signal resembles a single photon.

In this analysis we focus on the invisible and prompt-resolved signatures. We make this choice because the displaced-resolved and merged signatures require more precise detector and background modeling, which is beyond the scope of this project, while the invisible and prompt-resolved signatures give a clean first projection. The decay width, decay length, decay probabilities, and angular separation derived above are the quantities used to make this classification quantitative.

3 Simulation and Detector Modeling

3.1 Simulation strategy

The analysis is a counting problem repeated over the $(m_a, g_{a\gamma\gamma})$ plane. For each point we ask two things. First: what kind of event would an ALP make in the detector? Second:

would that event be large enough compared with the Standard Model background to be excluded? The first question is mostly analytic. The second needs simulation.

The basic event yield is

$$N = \sigma \mathcal{L}, \quad (55)$$

where σ is the cross section and $\mathcal{L}$ is the integrated luminosity. For the FCC-ee Z-pole run we use $\mathcal{L} = 150 \text{ ab}^{-1} = 1.5 \times 10^8 \text{ pb}^{-1}$, so even very small cross sections can give visible event counts. But Eq. (55) is not yet what the detector sees. A produced event still has to decay in the right part of the detector, pass the photon acceptance, be reconstructed, and fall into the analysis histogram. This is why we use

$$N_{\text{sig}} = \mathcal{L} \sigma(m_a, g_{a\gamma\gamma}) P_{\text{region}}(m_a, g_{a\gamma\gamma}) \epsilon_{\text{param}} C_{\text{Delphes}}. \quad (56)$$

Here P_{region} is the probability that the ALP is in the signal region, ϵ_{param} is the simple photon acceptance and reconstruction efficiency, and C_{Delphes} is the correction measured from detector-level simulation.

This is the central logic of the project. The analytic formulas tell us where the ALP decays, what the recoil energy is, and how the yield scales with $g_{a\gamma\gamma}$. Delphes tells us what survives as reconstructed photons. The Standard Model simulation tells us how much background sits under the same observable. The exclusion contour is then the place where the expected signal equals the signal required to stand out from that background.

We do not run the full detector simulation at every grid point. That would be slow and mostly unnecessary, because for associated production $\sigma(e^+e^- \rightarrow \gamma a) \propto g_{a\gamma\gamma}^2$ exactly at tree level. So the dense scan is analytic, while the full MadGraph–Pythia–Delphes chain is used to validate and correct it. The final inputs are:

- an analytic grid of $180 \times 180 = 32\,400$ points;
- 284 detector-level ALP signal points, each with 10 000 events;
- one $e^+e^- \rightarrow \gamma\gamma\gamma$ background sample with 10 000 events;
- one $e^+e^- \rightarrow \gamma\nu\bar{\nu}$ background sample with 10 000 events.

The generated Monte Carlo event counts are not the FCC-ee event counts. They are finite samples used to estimate shapes, efficiencies, and checks. The physical normalization always comes from the MadGraph cross section multiplied by the FCC-ee luminosity.

3.2 Scan ranges and model choices

The final FCC-ee projection is evaluated on a logarithmic 180×180 grid,

$$m_a \in [10^{-2}, 80] \text{ GeV}, \quad g_{a\gamma\gamma} \in [10^{-8}, 10^{-1}] \text{ GeV}^{-1}. \quad (57)$$

The lower mass bound is chosen because below about 10 MeV the two photons from $a \rightarrow \gamma\gamma$ are too collimated to be resolved at the Z pole for the angular resolution used here. The upper mass bound is chosen because associated production is strongly suppressed close to the kinematic endpoint, through the factor $(1 - m_a^2/s)^3$. The upper value 80 GeV still keeps the high-mass collider region, but avoids turning the result into a threshold study. The coupling range is wide on purpose. The lower end is below the event-yield floor at 150 ab^{-1} , while the upper end brackets the lifetime boundary where the ALP decays before it can escape the detector. The very large coupling end is therefore a numerical bracket, not a claim that every point is a controlled EFT benchmark.

We focus on the FCC-ee Z-pole run because it has the largest projected luminosity, about 150 ab^{-1} , and therefore gives the cleanest first benchmark for a rare process. The FCC-ee will also run at higher energies, but those stages have smaller luminosities and would require separately optimized searches. In this project the Z pole is used only as a high-luminosity e^+e^- collider environment. We do not use resonant $Z \rightarrow \gamma a$ production. Instead, we keep the photophilic benchmark in which the ALP couples to photons and is produced by ALP-strahlung, $e^+e^- \rightarrow \gamma^* \rightarrow \gamma a$.

There is one technical point in connecting this physical benchmark to the UFO model. The imported UFO does not expose $g_{a\gamma\gamma}$ directly. It uses the internal parameters f_a , K_B , and K_W , which are related to the physical coupling by

$$g_{a\gamma\gamma} = \frac{\alpha_{\text{em}}(K_B + K_W)}{\sqrt{2}\pi f_a}. \quad (58)$$

In the scan we fix $f_a = 1000 \text{ GeV}$ and vary the dimensionless sum $K_B + K_W$ to obtain the requested value of $g_{a\gamma\gamma}$. This is only a coordinate choice inside the UFO. The reported result is always in the physical $(m_a, g_{a\gamma\gamma})$ plane. With this choice,

$$K_B + K_W = \frac{\sqrt{2}\pi f_a}{\alpha_{\text{em}}} g_{a\gamma\gamma} = 6.09 \times 10^5 \left(\frac{g_{a\gamma\gamma}}{\text{GeV}^{-1}} \right). \quad (59)$$

Therefore the grid $g_{a\gamma\gamma} \in [10^{-8}, 10^{-1}] \text{ GeV}^{-1}$ corresponds internally to

$$K_B + K_W \in [6.09 \times 10^{-3}, 6.09 \times 10^4]. \quad (60)$$

The upper end is only used to bracket the short-lifetime boundary, and should not be interpreted as a statement that every point is a perturbative EFT model.

The UFO also contains an $a\gamma Z$ vertex. In the model this coupling is proportional to

$$C_{\gamma Z} \propto K_W c_w^2 - K_B s_w^2, \quad (61)$$

where s_w and c_w are computed from the same electroweak inputs as the UFO. To remove

the tree-level ALP– Z coupling while keeping the desired $a\gamma\gamma$ coupling, the parameter-card writer uses

$$K_B = (K_B + K_W)c_w^2, \quad K_W = (K_B + K_W)s_w^2. \quad (62)$$

This makes Eq. (61) vanish at tree level. Numerically, with the electroweak inputs used by the UFO, $s_w^2 = 0.212154$ and $c_w^2 = 0.787846$, so the code sets $K_B = 0.787846(K_B + K_W)$ and $K_W = 0.212154(K_B + K_W)$ for every generated point. This is important because otherwise the Z -pole result would mix two different questions: photophilic ALP-strahlung and resonant $Z \rightarrow \gamma a$ production.

3.3 MadGraph

The detector-level chain for one signal point is:

$$\begin{aligned} \text{param_card} &\rightarrow \text{MADGRAPH (LHE)} \rightarrow \text{PYTHIA (HepMC)} \\ &\rightarrow \text{DELPHES (ROOT)} \rightarrow \text{Python validation.} \end{aligned} \quad (63)$$

MadGraph is the first simulation step. It takes the UFO model, the beam energy, the process definition, and the parameter card for one $(m_a, g_{a\gamma\gamma})$ point. For the signal the process is $e^+e^- \rightarrow \gamma a$. MadGraph computes the matrix element, integrates the phase space, reports the production cross section, and writes an LHE event file. The LHE file is still a generator-level file: it contains the hard scattering particles and momenta, but it does not yet contain the detector response.

The same MadGraph step is also used for the Standard Model backgrounds. The resolved-prompt background is $e^+e^- \rightarrow \gamma\gamma\gamma$, and the invisible background is $e^+e^- \rightarrow \gamma\nu\bar{\nu}$. These backgrounds are generated separately from the signal because the background does not depend on m_a or $g_{a\gamma\gamma}$. Once the background shapes are made, they can be reused for every ALP point.

3.4 MadGraph verification

The first validation is a cross-section check. MadGraph must reproduce the analytic associated-production result,

$$\sigma(e^+e^- \rightarrow \gamma a) = \frac{\alpha g_{a\gamma\gamma}^2}{12} \left(1 - \frac{m_a^2}{s}\right)^3, \quad (64)$$

after converting from GeV^{-2} to pb. This test checks that the UFO coupling, the process definition, the beam setup, and the parameter-card mapping are all consistent with the analytic convention used in the theory section. In the full FCC-ee signal campaign all 284 detector-level ALP points passed this Gate 1 check. A separate Standard Model $e^+e^- \rightarrow \mu^+\mu^-$ smoke test was also used before the ALP scan, because it is a simple

way to check that MadGraph produces sensible e^+e^- events at the chosen center-of-mass energy.

3.5 Pythia

Pythia reads the LHE file from MadGraph. In this project it is used for two things. First, it adds QED radiation to the event record. Second, for the full ALP detector-level samples, it decays the ALP to two photons with the physical lifetime. The lifetime is obtained from the width

$$\Gamma(a \rightarrow \gamma\gamma) = \frac{g_{a\gamma\gamma}^2 m_a^3}{64\pi}. \quad (65)$$

Pythia needs the lifetime in millimeters, while the analytic calculation uses meters, so this is a place where a unit mistake would immediately change the signature classification. After Pythia, the event is written as a HepMC file, which is the format passed to Delphes.

3.6 Pythia verification

The Pythia validation checks the width and lifetime convention. The project uses the 64π convention in Eq. (65), and the parameter card writer uses the same convention when it writes the ALP width. The Pythia stage then checks that the decay length implied by the event record is consistent with the analytic boosted decay length,

$$\ell_a = \frac{|\mathbf{p}_a|}{m_a} \frac{\hbar c}{\Gamma_a}. \quad (66)$$

This is Gate 2. It matters because the invisible and prompt-resolved contours are lifetime boundaries as much as cross-section boundaries. If the ALP lifetime were off by a factor of two, the invisible and prompt regions would move in the $(m_a, g_{a\gamma\gamma})$ plane.

3.7 Delphes

Delphes is not a full detector simulation, but it is enough for this project because we need the main reconstruction effects: acceptance, photon efficiency, energy smearing, object counting, and missing energy. The locked detector numbers are:

- Photon acceptance: $|\eta| < 3.0$, $E_\gamma > 0.5 \text{ GeV}$;
- Photon reconstruction efficiency: 99%;
- Diphoton angular resolution: $\Delta\theta_{\text{res}} = 1.5^\circ$ (conservative calorimeter-cell scale, $\pi/120$);
- Tracker inner radius (L_{min}): 0.02 m;
- Tracker/calorimeter outer radius (L_{max}): 2.5 m.

For the FCC-ee projection we use an IDEA-like Delphes card. The Delphes output is a ROOT file with reconstructed objects such as photons and missing transverse energy. The Python analysis then reads these ROOT files with `uproot`, forms the diphoton invariant mass for the prompt-resolved channel, and forms the recoil-photon energy for the invisible channel.

The background samples are passed through the same MadGraph–Pythia–Delphes chain at $\sqrt{s} = 91.2 \text{ GeV}$. Two channels are generated:

- **Resolved-prompt background:** $e^+e^- \rightarrow \gamma\gamma\gamma$ (SM QED triple-photon production), with $\sigma = 7.306 \text{ pb}$ and $N_{\text{gen}} = 10\,000$ events. After Delphes and object selection, the resolved histogram contains 23 592 diphoton-pair entries. This number can be larger than the number of selected events because one event with several photons contributes several photon pairs.
- **Invisible background:** $e^+e^- \rightarrow \gamma\nu\bar{\nu}$ (radiative neutrino production), with $\sigma = 134.9 \text{ pb}$ and $N_{\text{gen}} = 10\,000$ events. After Delphes and the exactly-one-photon selection, the recoil-energy histogram contains 2 684 entries.

The backgrounds are generated separately from the signal. This is much more practical than generating a new mixed sample for every possible ALP mass and coupling. For the limit, each background histogram is normalized to the FCC-ee luminosity,

$$B_i = n_i^{\text{raw}} \frac{\sigma_{\text{bkg}} \mathcal{L}}{N_{\text{gen}}}, \quad (67)$$

where n_i^{raw} is the raw bin count in the Monte Carlo sample. The background does not change the ALP lifetime or topology. It only changes the number of signal events needed to claim sensitivity.

3.8 Delphes verification

The Delphes verification checks that the detector-level ROOT files contain the objects needed by the analysis and that the reconstructed observables peak in the expected places. For prompt-resolved ALP events, the relevant object is the best reconstructed photon pair in events with at least three photons. Its invariant mass must peak near the generated m_a . For invisible events, the recoil-photon energy must peak near the two-body value $E_\gamma = (s - m_a^2)/(2\sqrt{s})$. In the full channel-aware campaign, all 284 detector-level signal points passed the detector validation and were used to build the correction map C_{Delphes} in Eq. (56).

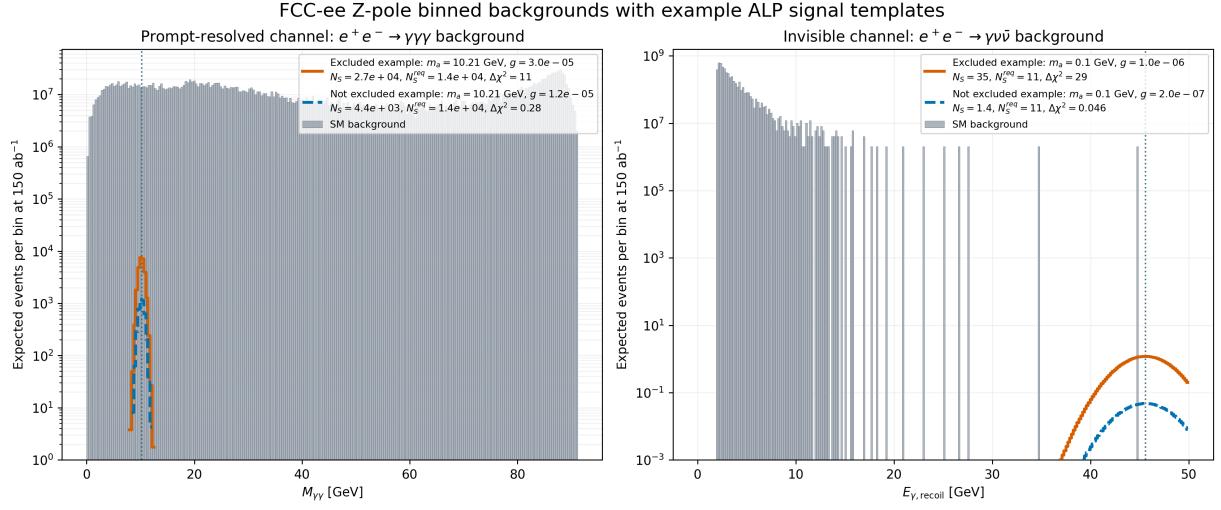


Figure 3: Background histograms used in the FCC-ee limit calculation, with example ALP signals overlaid. The resolved channel uses $M_{\gamma\gamma}$, while the invisible channel uses recoil-photon energy.

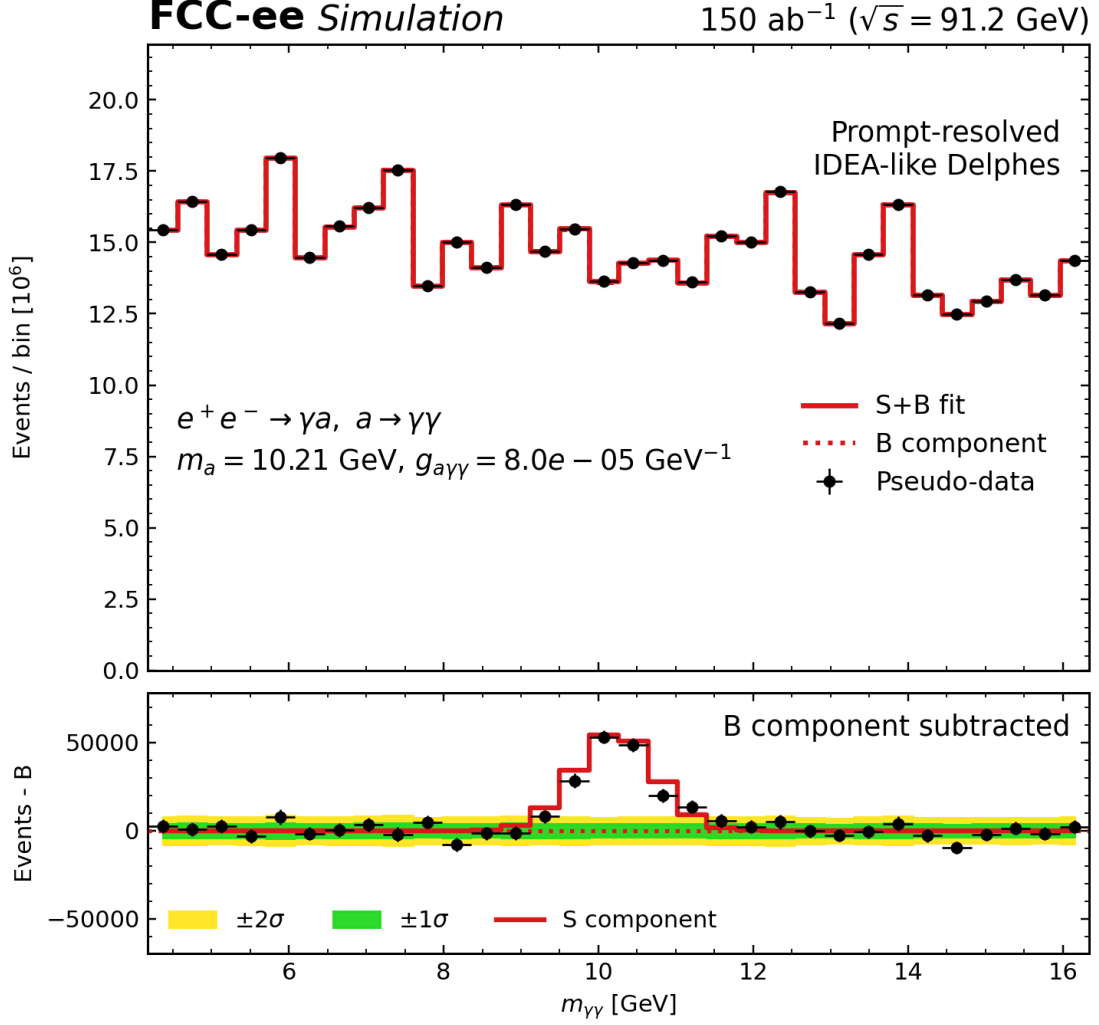


Figure 4: Prompt-resolved invariant-mass verification plot for one example ALP point, $m_a = 10.21$ GeV and $g_{a\gamma\gamma} = 8.0 \times 10^{-5}$ GeV⁻¹. The top panel shows pseudo-data, the Standard Model background template, and the signal-plus-background template. The lower panel subtracts the background, making the reconstructed ALP peak visible relative to the expected background fluctuations.

3.9 Belle II closure test

The final validation is the Belle II closure test. This is different from the previous checks because it tests the physics interpretation, not just whether the software chain runs. The code loads the public Belle II contour from AxionLimits, converts it to the same units, applies the same analytic production, lifetime, acceptance, and prompt-resolved logic used later for FCC-ee, and checks whether the contour can be reconstructed. The maximum residual is

$$\max_{m_a} \left| \log_{10} \left(\frac{g_{\text{closure}}}{g_{\text{published}}} \right) \right| = 7.6 \times 10^{-3}, \quad (68)$$

so the public-contour closure passes well within the 2% tolerance.

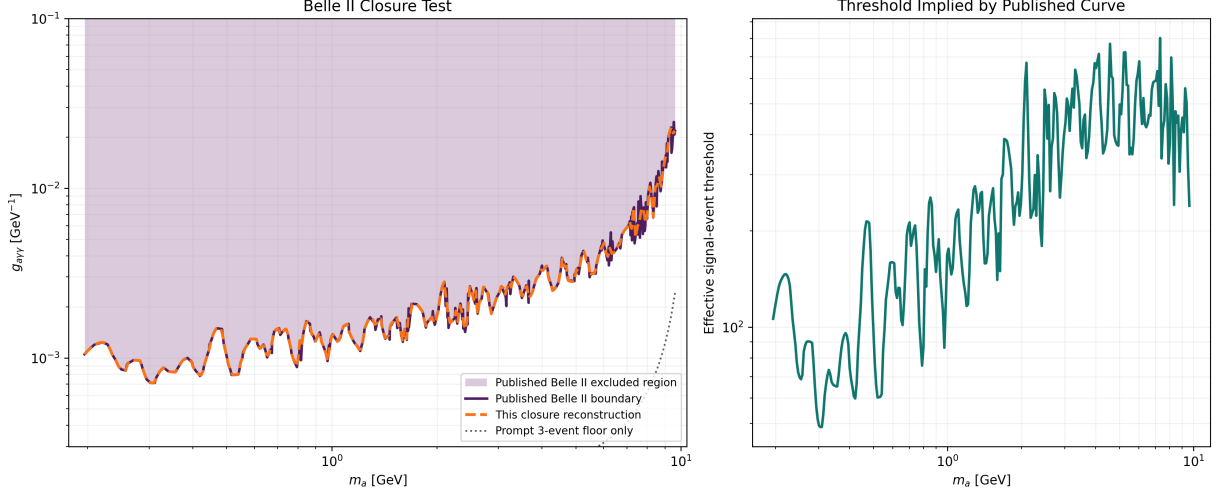


Figure 5: Belle II public-contour closure test. The reconstructed contour agrees with the published Belle II boundary, which checks the cross-section, lifetime, acceptance, and prompt-resolved logic used later for FCC-ee.

4 Analysis

4.1 Signature classification and efficiency

Each of the $180 \times 180 = 32\,400$ grid points $(m_a, g_{a\gamma\gamma})$ is classified as invisible, prompt-resolved, displaced-resolved, or merged. This is not yet an exclusion limit. It is just the map of what kind of signal the detector would see. The code compares the decay length ℓ_a with $L_{\min}$ and $L_{\max}$, and compares the photon opening angle with $\Delta\theta_{\text{res}}$. The prompt-resolved threshold mass is set by $\Delta\theta_{\min} = \Delta\theta_{\text{res}}$:

$$m_a^{\text{thresh}} = \frac{\sqrt{s} \Delta\theta_{\text{res}}}{4} \approx \frac{91.2 \text{ GeV} \times 0.0262}{4} \approx 0.597 \text{ GeV}. \quad (69)$$

Below this value the photons are merged. Above it, prompt decays can be used in the resolved three-photon search.

The lifetime probabilities used in Eq. (56) are

$$P_{\text{invisible}} = e^{-L_{\max}/\ell_a}, \quad (70)$$

$$P_{\text{prompt}} = 1 - e^{-L_{\min}/\ell_a}. \quad (71)$$

The detector correction is obtained from the 284 full signal points. These points were chosen on the contour-relevant branches: the lower invisible branch, the upper invisible lifetime branch, and the prompt-resolved branch. For each point the analysis counts how many generated signal events become usable reconstructed events. The ratio to the

analytic efficiency is

$$C_{\text{Delphes}} = \frac{f_{\text{selected}}^{\text{Delphes}}}{P_{\text{region}} \epsilon_{\text{param}}}. \quad (72)$$

This correction is then interpolated branch by branch and put back into Eq. (56). The prompt-resolved and lower invisible branches have corrections close to one. The upper invisible branch is less stable because it lies on a steep lifetime boundary, where small changes in lifetime produce very large changes in survival probability.

4.2 Limit-setting method

A projected 90% confidence level exclusion is calculated with a binned Asimov $\Delta\chi^2$ test [16]. “Asimov” means that the expected data are assumed to be exactly the Standard Model background. So the question is: if FCC-ee saw no excess, what ALP signal would be large enough to exclude?

The resolved channel uses the diphoton invariant mass $M_{\gamma\gamma}$, with a Gaussian signal peak centered at m_a . The invisible channel uses the recoil photon energy, with a Gaussian peak centered at $E_\gamma = (s - m_a^2)/(2\sqrt{s})$. In both cases the signal template is written as bin fractions f_i , while the background expectation in each bin is B_i .

With background-only Asimov data, the binned test statistic reduces to

$$\Delta\chi^2(N_{\text{sig}}) = \sum_i \frac{(N_{\text{sig}} f_i)^2}{\max(B_i, 1)}. \quad (73)$$

The denominator uses $\max(B_i, 1)$ so that empty Monte Carlo bins do not give infinite sensitivity. Solving $\Delta\chi^2 = 2.71$ gives the required signal count:

$$N_{\text{sig}}^{\text{req}} = \max \left(3, \sqrt{\frac{2.71}{\sum_i f_i^2 / \max(B_i, 1)}} \right), \quad (74)$$

The factor of 3 is a conservative minimum-event floor. The contour is then found by solving

$$N_{\text{sig}}(m_a, g_{a\gamma\gamma}) = N_{\text{sig}}^{\text{req}}(m_a) \quad (75)$$

at each mass. If the expected signal is above the required signal, that point would be excluded in the projected background-only FCC-ee dataset. The invisible contour has two branches because its yield is non-monotonic in $g_{a\gamma\gamma}$: at small coupling too few ALPs are produced, while at large coupling the ALP decays before escaping.

5 Results

Figure 6 shows the topology map. The invisible region dominates at low coupling, the prompt-resolved region appears at higher coupling above the angular-resolution threshold, the merged region sits at low mass where the two photons are too close, and the displaced region is the transition between prompt and invisible. This is useful because it tells us which analysis should be used before we make the exclusion contour.

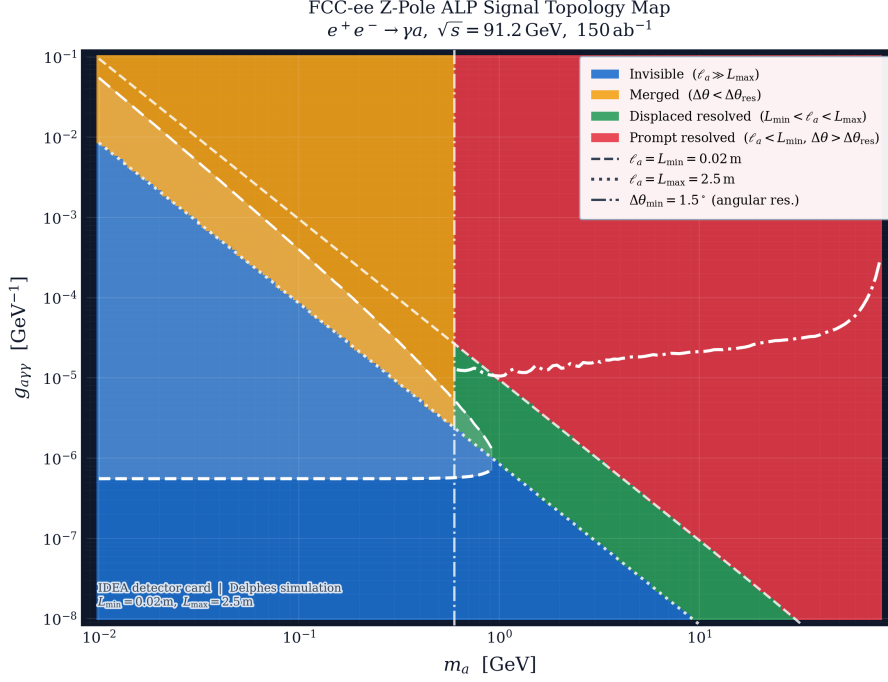


Figure 6: ALP signature classification in the $(m_a, g_{a\gamma\gamma})$ plane for the FCC-ee Z pole ($\sqrt{s} = 91.2 \text{ GeV}$) under IDEA detector assumptions. Dashed lines show the analytic $\ell_a = L_{\min}$ (prompt/displaced boundary) and $\ell_a = L_{\max}$ (displaced/invisible boundary) curves. The thin dash-dot curve shows the angular-resolution boundary $\Delta\theta = \Delta\theta_{\text{res}} = 1.5^\circ$. Only the invisible and prompt-resolved signatures are used in the projected exclusion contours.

Figure 7 shows the main result: the projected FCC-ee Z-pole exclusion contours overlaid on the current ALP-photon constraint landscape from AxionLimits [12].

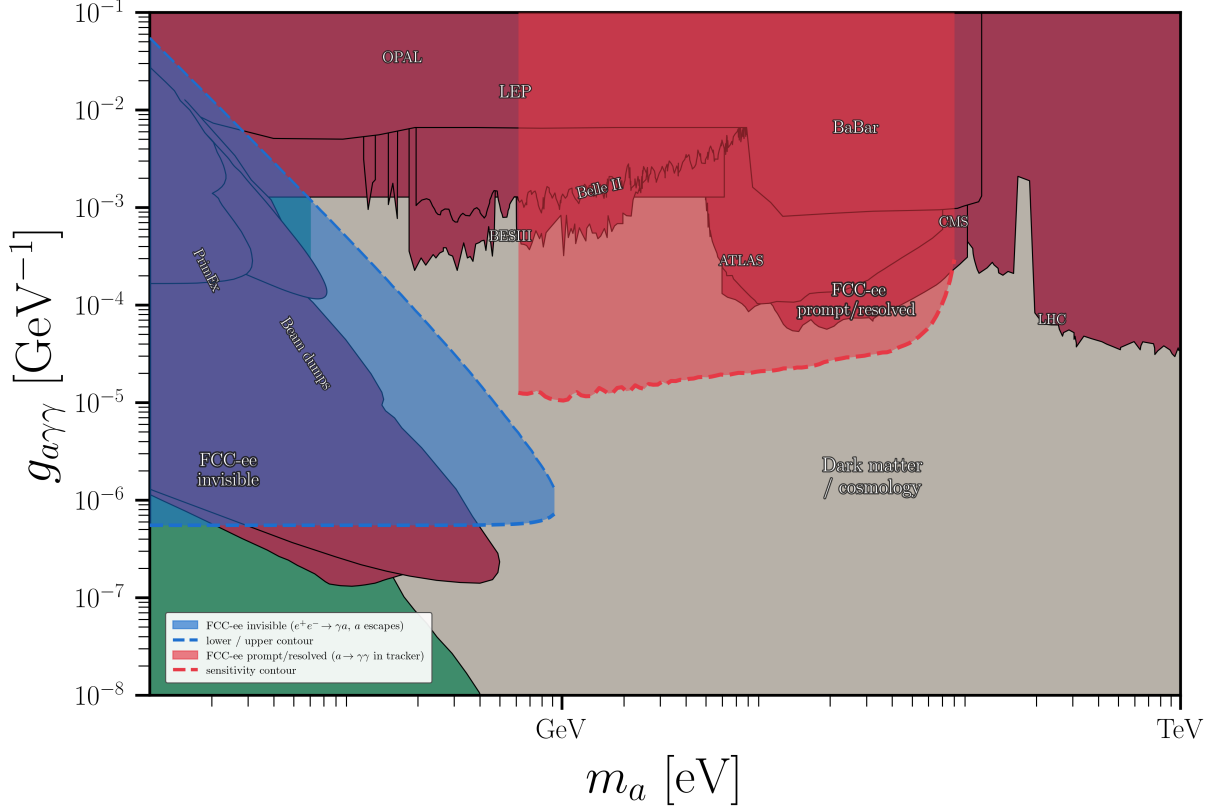


Figure 7: Detector-relevant ALP-photon parameter space with FCC-ee Z-pole projected sensitivity. Existing constraints are shown as solid colored regions from AxionLimits [12]. The translucent blue band is the FCC-ee invisible projection, bounded by the lower sensitivity contour and the upper short-lifetime contour. The translucent red region is the FCC-ee prompt-resolved projection, bounded by the dashed sensitivity contour.

The invisible channel reaches $g_{a\gamma\gamma} \simeq 5.5 \times 10^{-7} \text{ GeV}^{-1}$ for $m_a = 0.01\text{--}0.92 \text{ GeV}$. It has a lower sensitivity branch and an upper lifetime branch. The prompt-resolved channel starts at $m_a \simeq 0.61 \text{ GeV}$, set by the angular-resolution threshold, and extends to 80 GeV . Its contour is roughly $g_{a\gamma\gamma} \sim 10^{-5}\text{--}3 \times 10^{-4} \text{ GeV}^{-1}$ across the mass range. In the $1\text{--}10 \text{ GeV}$ region this is about one to two orders of magnitude below current Belle II and BaBar bounds.

The external AxionLimits regions are used as context, not as one single statistically identical exclusion. They include laboratory searches, astrophysics, cosmology, dark matter assumptions, and QCD-axion model bands. Our FCC-ee contour is a direct collider projection for the photophilic benchmark only.

6 Discussion

The main strength of the result is that the same logic is checked at several levels. The production cross section is analytic and verified with MadGraph. The width and lifetime are fixed to the same convention in theory, Pythia, and the parameter card. The Belle II

closure checks that the public contour can be reconstructed. The FCC-ee signal points check that the final reconstructed observables are where they should be: $M_{\gamma\gamma} \simeq m_a$ for prompt-resolved events, and the recoil photon energy near $(s - m_a^2)/(2\sqrt{s})$ for invisible events.

The result should still be read as a projection, not as an experimental limit. The prompt-resolved branch and the lower invisible branch are the most robust, because their detector corrections are close to one and their observables are simple. The upper invisible branch is less precise because it is a lifetime ceiling: the survival probability changes very quickly with $g_{a\gamma\gamma}$, so small detector or ISR effects can move the contour.

The main missing pieces are also clear. We do not include machine backgrounds, full detector systematics, reducible backgrounds from misidentified objects, or higher-order corrections to the SM background rates. We also do not use the merged and displaced signatures in the final limits. Those regions are real and visible in the topology map, but they would need dedicated cluster-shape and displaced-vertex studies. Finally, the present study cancels the tree-level ALP- Z coupling. Adding it would be a different two-coupling project, with resonant $Z \rightarrow \gamma a$ production at the Z pole.

7 Conclusion

We have computed projected 90% CL exclusion contours for photophilic ALPs at the FCC-ee Z pole, using analytic production and lifetime formulas corrected with a MadGraph-Pythia-Delphes simulation pipeline. The invisible channel reaches about $g_{a\gamma\gamma} = 5.5 \times 10^{-7} \text{ GeV}^{-1}$ for $m_a = 0.01\text{--}0.92 \text{ GeV}$. The prompt-resolved channel covers $m_a = 0.61\text{--}80 \text{ GeV}$ with sensitivity around $10^{-5}\text{--}3 \times 10^{-4} \text{ GeV}^{-1}$. The Belle II public-contour closure passes, and the FCC-ee pipeline includes detector-level signal checks and SM background histograms. The main deliverable is therefore a complete and clone-ready ALP simulation and projection pipeline, with a clear path to extend the study to merged, displaced, and ALP- Z signatures.

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Appendix: Contribution Statement

Iñaki Pardo led the physics study, repository organization, simulation campaigns, analysis development, validation, plotting, and writing. The project was developed in collaboration with project partners and with Codex-assisted implementation and documentation work. External software, models, detector cards, constraint data, and papers are credited in the repository provenance report and bibliography.